



Hack-Nation

In collaboration with MIT Club of Northern California and MIT Club of Germany

CHALLENGE

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The VC Brain:

Deploying \$100K Checks in 24 Hours

POWERED BY

MASCHMEYER GROUP



Maschmeyer Group — Investing in Exceptional Founders

1. Motivation / Goal to Achieve

Imagine running the **world's largest Shark Tank for AI innovation**. Thousands of ambitious builders, any one of whom could be building the **next Cursor, whose founders met at an MIT hackathon** before anyone knew to look. Your job: find them first, understand what they're capable of, and back them before the rest of the world catches on.

Right now, that's **nearly impossible**. Founders stay invisible until they know the right person. Their story is scattered across pitch decks, GitHub repos, half-built websites, and social posts nobody's reading closely. Diligence takes weeks. Capital flows through networks, not merit. By the time a fund finally sees a founder clearly, dozens of equally strong ones have already given up waiting. Great ideas are dying not for lack of merit, but for lack of a fair shot at being seen.

The fix has to work both ways. Some founders will be spotted first — for example through a GitHub commit, a hackathon win, a paper worth a phone call. Others won't wait to be discovered: a student with an idea, a first-time founder with zero connections, an engineer who's never pitched anyone in their life, simply applying. Either way, the same thing happens next: within 24 hours, they know if they just got \$100K to build it. Not because of who they know, but because of what the system already knows about them — a living profile of skills, experience, and track record that gets sharper with every milestone — a credit score for founders, the Founder Score. This is not just a narrative device: your system should actually produce this score, and use it as one input into every investment decision. Ship something once, and your next idea starts from a stronger position. The system never forgets, and it never stops updating.

And you're not an outside observer here — most of you are exactly the founder profile this system is built to find. Build it well, and you'll understand better than almost anyone what actually gets noticed, and walk away with the system you wish had found you.

Ambitious Goal: Build a data- and AI-first operating system that transforms how venture capital works — discovering exceptional founders, deploying capital at 100× speed, and supporting the next generation of AI entrepreneurs. Identify and track potential founders and teams, deploying \$100K checks within 24 hours.

Done right, this doesn't just make venture capital faster — it makes it accessible: a founder's potential becomes something the system can see and reason about directly.

The North Star — a fund that runs itself: The ultimate vision is an autonomous venture fund spanning the entire lifecycle, with one human in the loop for oversight, not execution. This challenge covers Sourcing → Screening → Diligence → Decision. Downstream stages (portfolio monitoring, follow-on, fund ops, exit) are real but out of scope — don't spend hackathon time designing UI for them.

2. Build the VC Brain (MVP)

The VC Brain is built on three transformative pillars that work in concert to give a single investor the reach and analytical power of an entire organisation.

Sourcing — how do you find the best founders? The most important part of your MVP.

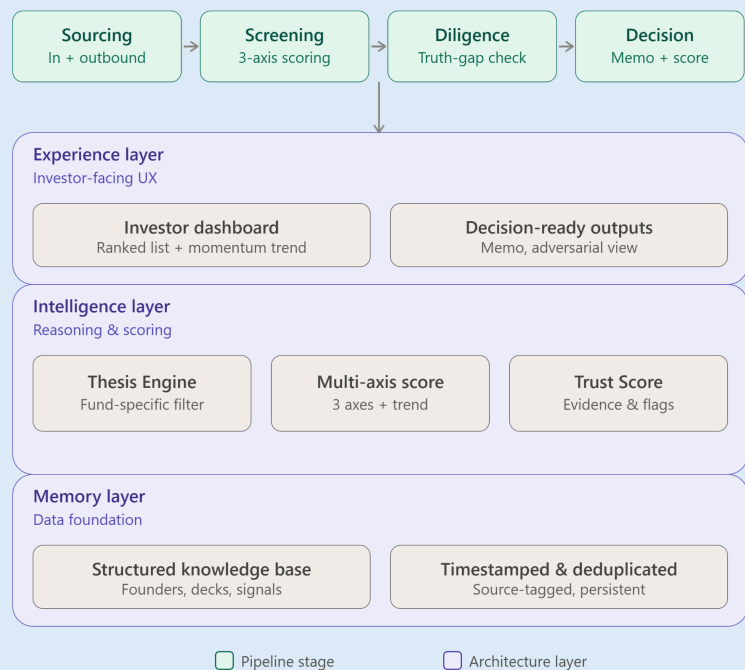
- Surface the strongest founders before they formally begin fundraising.
- Judged on data richness and smart sourcing ideas, not polish.
- Least commercial competition today — go further here than anywhere else.

Assessment & Intelligence — the reasoning layer. Makes or supports every investment decision.

- Operates on top of Memory to produce insights, challenge assumptions, and recommend next steps.
- Triggered by an inbound application, or by signals crossing a conviction threshold on their own.
- Transparent about confidence, uncertainty, and the evidence behind every conclusion.

Memory — the data foundation. Nothing discarded.

- Ingests pitch decks, interviews, launches, GitHub activity, and social traction.
- Deduplicates, enriches, timestamps, and tags everything by source.
- Houses the Founder Score — persists across applications, never resets.
- Surfaces the trend over time, not just the latest snapshot.



VC Brain architecture: pipeline stages (Sourcing → Screening → Diligence → Decision) mapped onto the Memory, Intelligence, and Experience layers.

The MVP should demonstrate:

- 1. Thesis Engine:** Investor sets sectors, stage, geography, check size, ownership targets, and risk appetite. Every recommendation is filtered and scored through this fund-specific lens.
- 2. Smart Data Collection & Management:** Actively collect, validate, and structure founder and company data from heterogeneous sources — the data layer matters as much as the intelligence built on top of it.
- 3. Multi-Attribute Reasoning:** Move beyond keyword search. Support complex, natural-language queries, e.g. “technical founder, Berlin, AI infra, enterprise traction, no prior VC backing, top-tier accelerator.”

4. Inbound: Application & Automated Screening

- **Apply:** deck + company name is the minimum bar; any further fields are the minimum needed for a confident 24-hour decision.
- **Screen:** a fast first-pass filter removes clearly non-viable ideas before full analysis begins.

5. Outbound: Founder Identification & Activation

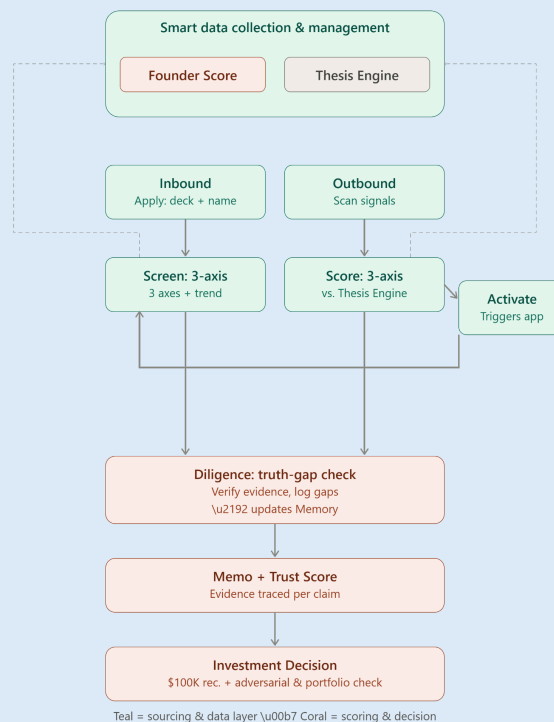
- **Identify:** continuously scan GitHub, launches, hackathons, papers / patents, and accelerator cohorts — scored the same way as an inbound application.
- **Activate:** reach out to the strongest matches directly. Cold outreach, not cold investment — the goal is to trigger a real application.
- **Converge:** activated applications flow into the same Screening step as inbound, so both tracks feed one funnel.

6. Multi-Axis Screening: Every opportunity is scored along three independent axes — not averaged:

- **Founder:** who they are, their traits and track record.
- **Market:** sizing, competitors, SWOT — rated bullish, neutral, or bear.
- **Idea vs. Market:** does the idea survive scrutiny as-is, or is the team strong enough to pivot?
- Each axis also shows trend (improving / declining / stable) and feeds back into Memory to sharpen future scoring.

7. Evidence-Backed Investment Memos & Trust Score: Every claim — traction, revenue, team background, market size — must trace to evidence with a confidence level: a Trust Score. Verify externally where possible and flag contradictions before they reach the investor.

8. Investor-Grade UX: Intuitive enough to use without technical support. Notion-level approachability, Bloomberg-level analytical depth — clarity and usability are non-negotiable.



End-to-end flow from inbound and outbound sourcing through 3-axis screening, diligence, and the final investment decision.

3. Stretch Goals

We know that you can do it!

- 1. Agentic Traceability:** Every recommendation must cite the exact data point — pitch deck slide, web signal, or interview excerpt — that drove the conclusion. If the system recommends a founder, it must show precisely what justified the Trust Score.
Hint: implement step-level chain-of-thought logging to visualise the full reasoning process.
- 2. Self-Correction Loops:** Implement a Validator Agent that cross-references extracted founder claims against market databases, comparable funding rounds, and observable evidence to ensure the primary agent is not hallucinating.
- 3. Sourcing & Network Intelligence:** Model the sourcing graph — the network of programs, institutions, and individuals through which promising founders become visible — and track which channels historically produce the strongest opportunities. Proactively suggest underexplored sourcing channels based on what has worked before, and once a founder converts into a funded deal, feed that outcome back into the model so it learns which channels generate quality, not just volume.

4. Areas of Research

Genuinely open problems. Solving them robustly could be industry-defining — document your approach if you crack one.

- 1. Confidence Scoring:** Founder data is messy and incomplete. Can prediction intervals be built around soft-skill assessments like resilience or founder-market fit?

2. Data Quality vs. Volume: More data isn't always better — how do you decide what's worth collecting vs. flagging as low-confidence?

3. Founder Traits & Success: How much can public footprints (Twitter, LinkedIn) predict founder success — and how would you test it?

5. Hints and Resources

Data

No dedicated dataset is provided. Bring or synthesise your own founder and company data — public web data (Crunchbase, LinkedIn, GitHub, ProductHunt, Hacker News, arXiv, patents), synthetic founder profiles with seeded contradictions, or anonymised / fictional pitch decks. Quality of ingestion matters more than dataset size.

Investment Memo — see Appendix 1: A sample checklist for the structure and depth expected — not a template to fill in mechanically.

6. Evaluation Criteria

Data Architecture and Intelligence (30%): How well does the system collect, structure, and manage founder data? We look for smart ingestion, deduplication, enrichment, and a reasoning layer that is honest about what it knows and what it does not.

***Note:** generic ingestion / enrichment quality alone will not score highly here if it doesn't address the cold-start, pre-track-record case.*

Intelligent Analysis and Trust (25%): How effectively does the solution synthesise fragmented signals into decision-ready insights? Do the memo's Trust Scores surface evidence and uncertainty transparently?

Investment Utility & Execution (30%): Does the tool produce a recommendation a human investor could genuinely act on within 24 hours? This includes any progress on instrumenting how fast and reliably an opportunity moves from first signal to decision. Could it transform the speed and quality of capital allocation?

User Experience and Design (15%): Is the interface intuitive, clear, and beautifully designed? Does it make complex AI reasoning feel effortless and trustworthy for a non-technical investor?

7. Why It Matters

Not just another investment tool: transformative infrastructure for the venture capital industry.

- Today, capital flows to who you know — not what you're building.
- A system that surfaces exceptional founders before anyone else sees them could reshape how the world's most important companies get their start.
- You are building infrastructure for Equitable Capital Allocation — turning a relationship-gated landscape into a living intelligence network that knows who the next Cursor is, and ensures they get funded.

Build the data layer. Build the reasoning layer. Build the experience. Build the infrastructure that gives exceptional founders the capital to begin.

Appendix 1 — Investment Memo

Length rule: As detailed as the decision requires, as brief as clarity allows. Length is not a proxy for rigor — padding counts against you.

How to use this checklist:

The sections below represent what a complete investment memo should cover. For this challenge, treat Company snapshot, Investment hypotheses, SWOT, Problem & product, and Traction & KPIs as required. Some sections — particularly Financials & round structure, Cap table, and anything involving internal company data — depend on information a real fund would treat as confidential or simply might not have at the time of drafting. You are not expected to fabricate this data convincingly. Where a data point is missing, unavailable, or intentionally left out, it must be explicitly flagged in the memo (e.g. “Cap table: not disclosed” or “Customer references: unavailable at this stage”) rather than silently omitted or guessed. A memo that clearly marks its own gaps is more trustworthy — and more useful to an investor — than one that fills them in invisibly.

Section	Should include
Company snapshot	One-paragraph “in a nutshell”: market size, the structural problem, why it’s urgent, and how the product solves it
Investment hypotheses	The explicit “why we want to invest” bullets — team quality, market wedge, stickiness / retention mechanics, traction signal, defensibility, expansion path
SWOT	Strengths, weaknesses, opportunities, risks — each as short, evidence-backed bullets
Team & history	Founder background, exec team pedigree, why the fund is comfortable with any red flags (e.g. single-founder), company timeline from founding to today
Problem & product	The core problem(s) in plain language, then the step-by-step product / process solving it
Technology & defensibility	What’s proprietary vs. commoditizable, the data moat, model / architecture choices, why the advantage compounds over time
Market sizing	Top-down and / or bottom-up TAM / SAM / SOM, with the assumptions stated explicitly
Competition	Named competitor clusters, how each differs from the company, and who could become a threat later
Traction & KPIs	Customer count, ARR / revenue, growth trajectory, unit economics (CAC, sales cycle, churn), usage metrics (e.g. DAU)
Financials & round structure	Historical + projected P&L (revenue, EBITDA, opex, COGS), round size, runway, next-round timing
Cap table	Pre- and post-round ownership by party, dilution assumptions, VSOP allocation
Due diligence log	What was checked (commercial, people, financial, legal, technical), what’s still open

Section	Should include
Exit perspective	Plausible exit paths (strategic acquirers, PE roll-up, category comparables) and why they'd pay a premium

FAQ

1. What does a winning sourcing layer actually do that a filtered list doesn't?

It surfaces founders before they start fundraising. Outbound scans GitHub, launches, hackathons, papers/patents, and accelerator cohorts, scores them like an inbound application, and triggers real outreach. The brief flags this as the area with the least commercial competition — go further here than anywhere else.

2. If we can only nail one thing, sourcing or the reasoning layer, which wins?

Sourcing carries the most weight (Data Architecture 30%) and is explicitly the priority. But a rich data layer with no honest reasoning on top scores poorly too. Build sourcing deep, then a thin-but-transparent Intelligence layer over it — not a polished reasoner over shallow data.

3. Is there a provided dataset?

No. Bring or synthesize your own — public web data (Crunchbase, LinkedIn, GitHub, ProductHunt, Hacker News, arXiv, patents), synthetic profiles with seeded contradictions, or fictional decks. Ingestion quality beats dataset size.

4. What's the minimum application input?

Deck + company name. Add fields only if genuinely needed for a confident 24-hour decision. Over-collecting works against you.

5. Can we just average the three screening axes into one number?

No — the brief says explicitly not averaged. Founder, Market, and Idea-vs-Market are scored independently, each with a trend direction, and each feeds the memo separately. Collapsing them hides exactly the disagreement an investor needs to see.

6. Founder Score vs. the 3-axis score — aren't these the same thing?

No. The Founder Score lives in Memory, persists across applications, and never resets — it follows the person across different startups over time. The 3-axis score is per-opportunity. The Founder Score is one input into the Founder axis, not a substitute for it.

7. Is the Trust Score one number for the company?

No — it's per claim. Each assertion (traction, revenue, team, market size) traces to evidence with a confidence level, verified externally where possible, with contradictions flagged before they reach the investor.

8. Which memo sections are actually required?

Company snapshot, Investment hypotheses, SWOT, Problem & product, Traction & KPIs. The rest are welcome but optional — and padding counts against you.

9. What do we do when real financials or cap table data don't exist?

Don't fabricate. Flag the gap explicitly ("Cap table: not disclosed"). A memo that marks its own gaps is scored as more trustworthy, not less.

10. How do you score a first-time founder with no GitHub, no funding, no network?

This is the cold-start case, and the brief warns that generic ingestion/enrichment won't score highly if it ignores it. You need an explicit method for pre-track-record founders — otherwise you've just rebuilt the network-gated system the challenge exists to replace. Don't leave this as an afterthought.

11. Is there a way to push the cold-start problem further, beyond just handling it competently?

Yes — worth attempting Area of Research 3: how much can public footprints (Twitter, LinkedIn) actually predict founder success, and how would you test it. It's framed as genuinely open, but it's also the most direct lever on the cold-start weakness in Q10 — a founder with no funding or GitHub history often still has a public footprint. Teams that take a real stab at this, even partially, are documenting exactly the kind of approach the brief says could be industry-defining.

12. How natural-language does Multi-Attribute Reasoning need to be?

Beyond keyword/filter search. It should resolve a compound query like "technical founder, Berlin, AI infra, enterprise traction, no prior VC backing, top-tier accelerator" in one pass — not as five manual filters.

13. Which stretch goal is worth it if we only have time for one?

Agentic Traceability — citing the exact data point (deck slide, web signal, interview line) behind each conclusion. It directly reinforces the core Trust Score requirement, so it's the highest-leverage add.

14. How much does UI polish matter under time pressure?

UX is 15% — real, but the smallest slice. "Notion-level approachability, Bloomberg-level analytical depth," usable without technical support. If forced to trade, protect the data and reasoning layers (55% combined) first.

15. Should the Thesis Engine be hardcoded to one fund, or configurable?

Configurable. The investor sets sectors, stage, geography, check size, ownership targets, and risk appetite, and every recommendation is filtered and scored through that lens. A hardcoded thesis misses the point of the pillar.

16. Do we need to build toward the "fund that runs itself"?

No — that's ambition framing, not a deliverable. The scope is Sourcing → Screening → Diligence → Decision. The brief explicitly says don't spend hackathon time on downstream stages (monitoring, follow-on, fund ops, exit).